

1. Administrative & Study Identification

Full Study Title A Phase 3 Randomized, Placebo-controlled Study to Evaluate the Safety and Efficacy of Pemetrexed + Platinum Chemotherapy + Pembrolizumab (MK-3475) With or Without Lenvatinib (E7080/MK-7902) as First-line Intervention in Participants With Metastatic Nonsquamous Non-small Cell Lung Cancer (LEAP-006)
Short Title/Acronym LEAP-006 **Protocol Number** MK-7902-006 / E7080-G000-315 **NCT Number** NCT03829319 **Posting ID** 779945756904293553
Trial Status COMPLETED **Sponsor Name** Merck **Investigational Product** Pembrolizumab (MK-3475) and Lenvatinib (MK-7902/E7080) plus Pemetrexed and Platinum Chemotherapy (Carboplatin or Cisplatin)
Phase of Development PHASE3 **Medical Monitor** Clinical Operations Lead, Merck

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate whether the combination of lenvatinib + platinum doublet chemotherapy + pembrolizumab prolongs Progression-Free Survival (PFS) and Overall Survival (OS) compared to matching placebo + platinum doublet chemotherapy + pembrolizumab as a first-line intervention, and to assess initial safety (Dose-limiting toxicities, AEs, and discontinuations due to AEs) in Part 1.
Secondary Objectives: To evaluate the Objective Response Rate (ORR), Duration of Response (DoR), safety, and quality of life (QoL) among the participants in Part 2.

2.2 Background & Rationale To address the need for improved first-line therapies in metastatic nonsquamous NSCLC patients without sensitizing genetic aberrations (EGFR, ALK, or ROS1). Lenvatinib, a multiple-receptor tyrosine kinase inhibitor, has demonstrated antitumor activity in combination with the anti-PD-1 inhibitor pembrolizumab or with chemotherapy in advanced NSCLC. This study was designed to ascertain if adding lenvatinib to the standard of care (pembrolizumab + chemotherapy) would yield superior survival outcomes.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled, Active Comparator **Blinding:** Double-blind (Part 2 efficacy phase); Open-label (Part 1 safety run-in)
Randomization: 1:1

3.2 Planned Enrollment & Duration Target **\$N\$:** 761 patients enrolled **Number of Sites:** 160 sites across 17 countries **Estimated Study Start:** March 25, 2019 **Estimated Primary Completion:** August 11, 2023 (Actual)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Histologically or cytologically confirmed metastatic (Stage IV, American Joint Committee on Cancer [AJCC] version 8 or current version) nonsquamous NSCLC. 3. Treatment-naïve in the metastatic setting (First-line). 4. Absence of tumor-activating EGFR mutations AND absence of ALK and ROS1 gene rearrangements, OR presence of a Kirsten Rat Sarcoma (KRAS) gene mutation. 5. Measurable disease per RECIST 1.1 and ECOG Performance Status of 0 or 1. 6. Life expectancy of at least 3 months. 7. Adequately controlled blood pressure (BP) defined as BP $\leq 150/90$ mm Hg. 8. Adequate organ function. 9. Contraceptive use by men and women per protocol guidelines.

4.2 Exclusion Criteria 1. Prior PD-1/PD-L1 therapy in the adjuvant setting. 2. Known untreated central nervous system (CNS) metastases and/or carcinomatous meningitis. 3. History of pneumonitis that required steroids, or active autoimmune disease. 4. Major blood vessel invasion or intratumoral cavitations. 5. Known positive status for HIV, active Hepatitis B (HBV), or Hepatitis C (HCV). 6. Active hemoptysis (at least 0.5 teaspoon of bright red blood) within 2 weeks. 7. Significant cardiovascular impairment within 12 months, or LVEF below institutional normal range.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Lenvatinib 8 mg daily (or matching placebo) + pembrolizumab 200 mg + pemetrexed 500 mg/m² + carboplatin AUC 5 mg/mL/min or cisplatin (Trade name: Platinol, ATC code: L01XA01) 75 mg/m² given every 3 weeks (Q3W) for 4 cycles. This is followed by maintenance pembrolizumab + lenvatinib/placebo + pemetrexed. **Route of Administration:** Oral (lenvatinib/placebo) and Intravenous (pembrolizumab, pemetrexed, carboplatin/cisplatin). **Duration of Treatment:** Pembrolizumab for up to 35 cycles (approximately 2 years); lenvatinib/placebo and pemetrexed until documented disease progression or intolerable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for chemotherapy-related toxicities. **Prohibited:** Concurrent investigational agents or other anti-cancer therapies. Prior PD-(L)1 therapy or EGFR/ALK/ROS1-directed therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Imaging, Genomic Testing
Induction	Cycles 1-4 (Q3W)	Randomization, IV Chemotherapy + Pembrolizumab + Oral Lenvatinib/Placebo
Maintenance	Cycle 5+ (Q3W)	Pembrolizumab + Pemetrexed + Lenvatinib/Placebo administration
Follow-Up	Q6W / Q9W / Q12W	Tumor imaging Q6W until wk 18; Q9W until wk 54; then Q12W

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. Part 1: Number of Participants With a Dose-limiting Toxicity (DLT). 2. Part 1: Number of Participants Who Experienced an Adverse Event (AE). 3. Part 1: Number of Participants Who Discontinued Study Drug Due to an Adverse Event. *(Study hypotheses also evaluate whether the combination prolongs Progression-Free Survival (PFS) and Overall Survival (OS) compared to matching placebo + platinum doublet chemotherapy + pembrolizumab).*

7.2 Secondary & Exploratory Endpoints 1. Part 2: Objective Response Rate (ORR) per RECIST 1.1 by BICR. 2. Part 2: Duration of Response (DoR) per RECIST 1.1 by BICR. 3. Part 2: Number of Participants Who Experienced an Adverse Event (AE). 4. Health-related Quality of Life (QoL).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: AEs were closely monitored and graded using NCI CTCAE v4.0 throughout the study duration.

Serious Adverse Events (SAEs): Standard protocol collection including immune-mediated AEs related to pembrolizumab and toxicities related to lenvatinib.

Data Monitoring Committee (DMC): An independent study oversight committee evaluated safety during Part 1 and ongoing efficacy/safety in Part 2.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy endpoints (PFS/OS). Safety Set for all participants who received at least one dose of study treatment.

Sample Size Justification: Target enrollment of 761 patients to ensure adequate power for dual primary endpoints of PFS and OS.

Handling of Missing Data: Handled per standard statistical guidelines for time-to-event endpoints; analyzed by Kaplan-Meier method and stratified log-rank test.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine clinical site monitoring, source data verification, and central laboratory utilization. **Audit Trail:** eCRF entry tracking, queries, and locked database procedures enforced by the sponsor.

11. Study Identifiers & Governance

Primary ID: MK-7902-006 / E7080-G000-315 **Registry Number:** NCT03829319 **Sponsor/Lead Organization:** Merck **Study Title:** LEAP-006 **Official Regulatory Title:** A Phase 3 Randomized, Placebo-controlled Study to Evaluate the Safety and Efficacy of Pemetrexed + Platinum Chemotherapy + Pembrolizumab (MK-3475) With or Without Lenvatinib (E7080/MK-7902) as First-line Intervention in Participants With Metastatic Nonsquamous Non-small Cell Lung Cancer (LEAP-006)

12. Clinical Framework (NSCLC Specific)

Disease Area: Nonsquamous Non-small Cell Lung Cancer (Preferred UMLS Name: Nonsquamous Non-Small Cell Lung Carcinoma) **Diagnosis Threshold:** Stage IV, EGFR-, ALK-, and ROS1-wild-type status confirmed (or KRAS mutant) **Medical Methodology:** PD-1 inhibitor + Platinum Doublet Chemotherapy $\pm$ Tyrosine Kinase Inhibitor **Optimization Target:** Prolongation of PFS and OS

13. Study Procedures & Methodology

Management Pathway: Standard oncology treatment pathway with added investigational TKI (Lenvatinib) **Education Protocol (HCP):** Investigator protocol training, AE management, and RECIST 1.1 evaluation training **Education Protocol (Patient):** Informed consent, oral medication diary, and AE self-reporting instructions **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) of all tumor imaging

14. Observation & Follow-Up Schedule

Total Duration: Minimum 24 months of treatment (or until disease progression), followed by survival follow-up phase. **Onsite Visit Cadence:** Every 3 weeks (Q3W) for IV infusions and safety checks. **Remote Monitoring Frequency:** Standard AE follow-up and survival status checks post-treatment. **Checklist Review Interval:** Tumor imaging every 6 weeks (Wks 1-18), every 9 weeks (Wks 18-54), and every 12 weeks thereafter.

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Insert Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Insert Name]	_____	[DD-MMM-YYYY]					
Lead Statistician	[Insert Name]	_____	[DD-MMM-YYYY]					